IT305PC: INTRODUCTION TO IOT

B.Tech. II Year I Sem.

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Course Objectives:

1. Understand the concepts of Internet of Things and able to build IoT applications
2. Learn the programming and use of Arduino and Raspberry Pi boards.
3. Know about data handling and analytics in SDN.

Course Outcomes: Upon completing this course, the student will be able to:

1. Understand key characteristics, layers, protocols, sensor networks and their role in IoT.
2. Differentiate between IoT and M2M communications and achieve interoperability using Arduino Programming.
3. Apply Python constructs on Raspberry Pi interfacing models and their use cases.
4. Design and implement SDN-based solutions for IoT deployments using Raspberry Pi
5. Apply the knowledge of cloud storage models to select an appropriate model for any IoT application.
6. Analyze different sensor technologies for real world applications through IoT.

UNIT- I

Introduction to IoT and Sensor Networks

Introduction to Internet of Things (IoT), Characteristics and Applications of IoT, IoT Architecture and Reference Models (IETF, ITU-T)
Physical Design of IoT: Devices, Gateways, and Data Centers
Functional Blocks of IoT: Sensing, Actuation, Communication, Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology
Basics of Networking and Communication Protocols: MQTT, CoAP, ZigBee,
HTTP Sensor Networks: Types, Topologies, and Protocols
Introduction to IoT Security and Privacy Fundamentals.

UNIT- II

Machine to Machine (M2M) and Embedded Programming for IoT

Machine-to-Machine Communications Overview, Difference between IoT and M2M, Interoperability in IoT: Standards and Protocols.
Introduction to Arduino Programming for IoT, Integration of Sensors and Actuators with Arduino. Hands-on Exercises: Sensor Data Acquisition and Actuator Control, Basic Communication Protocols Implementation on Arduino, IoT Device Interoperability Challenges and Solutions.

UNIT- III

Raspberry Pi with Python Programming for IoT

Introduction to Python Programming Basics
Overview of Raspberry Pi and Its Role in IoT, Interfacing Raspberry Pi with Sensors and Actuators (UART, SPI, I2C).
Data Acquisition and Local Processing on Raspberry Pi, Sending Data to Cloud Platforms.
Implementation of IoT Projects Using Raspberry Pi
Case Studies: Smart Home Automation, Healthcare Monitoring, Environmental Sensing

UNIT- IV

Network Management and Software Defined Networking (SDN) for IoT

Introduction to Software Defined Networking (SDN), SDN Architecture: Controller, Data Plane, Control Plane, Benefits of SDN in IoT: Flexibility, Centralized Control, Security, Traffic Management, SDN Models: Open SDN, API SDN, Overlay, Hybrid
Data Handling, Analytics, and Security Considerations in SDN for IoT

Network Management Challenges in IoT, Integration of SDN with IoT Networks

UNIT- V

Cloud and Edge Computing Models with IoT Use Cases

Introduction to Cloud Computing and Cloud Storage Models, Edge and Fog Computing Concepts for IoT, Web Servers and Cloud Platforms for IoT (AWS IoT, Azure IoT, etc.), Sensor-Cloud Architecture Data Analytics and Visualization Techniques for IoT Data
IoT Use Cases: Smart Cities, Smart Homes, Connected Vehicles, Smart Grid, Industrial IoT Case Studies: Agriculture, Healthcare, Activity Monitoring.

TEXT BOOKS

1. Pethuru Raj and Anupama C. Raman "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press.
2. Vijay Madisetti, Arshdeep Bahga, “Internet of Things: A Hands-On Approach”

REFERENCE BOOKS

1. Terokarvinen, kemo, karvinen and villeyvaltokari, “Make sensors”: 1st edition,2014, Maker media.
2. Waltenegus Dargie, Christian Poellabauer, Fundamentals of Wireless Sensor Networks: Theory and Practice.
3. Charles Bell, Beginning Sensor networks with Arduino and Raspberry Pi, 2013, Apress.
4. Fei Hu, Security and Privacy in Internet of Things (IoTs), CRC Press, Taylor & Francis Group.
5. S. Sahoo, S. Sahoo, S. Mishra, Software-Defined Networking for Future Internet Technology: Concepts and Applications, Routledge.
6. Honbo Zhou, The Internet of Things in the Cloud: A Middleware Perspective. 2012, CRC Press (Taylor & Francis).